



XXGRIDUTILITY CORP

PROJECT SENTINEL

Advanced Grid Telemetry & Forensic Analytics Architecture

Enterprise Infrastructure Audit & Resilience Strategy

Designed & Developed by Olinda Mascarenhas
Senior Technical Program Manager & Enterprise Infrastructure Leader



MACRO CONTEXT

The New Jersey Electrical Grid Footprint & Economic Impact

MARKET SCALE: NEW JERSEY

\$10B+

ANNUAL UTILITY REVENUE

The total revenue generated from electricity sales to NJ end-users, reflecting a highly industrial energy footprint.

20k

MEGAWATT PEAK LOAD (MW)

Extreme seasonal demand spikes driven by residential climate control and high-density industrial corridors.

¹ **Source:** U.S. Energy Information Administration (EIA) New Jersey Electricity Profile; Consolidated Annual 10-K Filings for PSEG, FirstEnergy (JCP&L), and Exelon (ACE).

RISING DEMAND & CONSTRAINTS

Data Center Surge

New Jersey is experiencing a "Philadelphia's worth" of new electricity users annually. Data centers now account for **63% of capacity price increases** in regional auctions.

This rapid load growth creates data entropy, making forensic visibility into telemetry dropouts a mission-critical requirement.

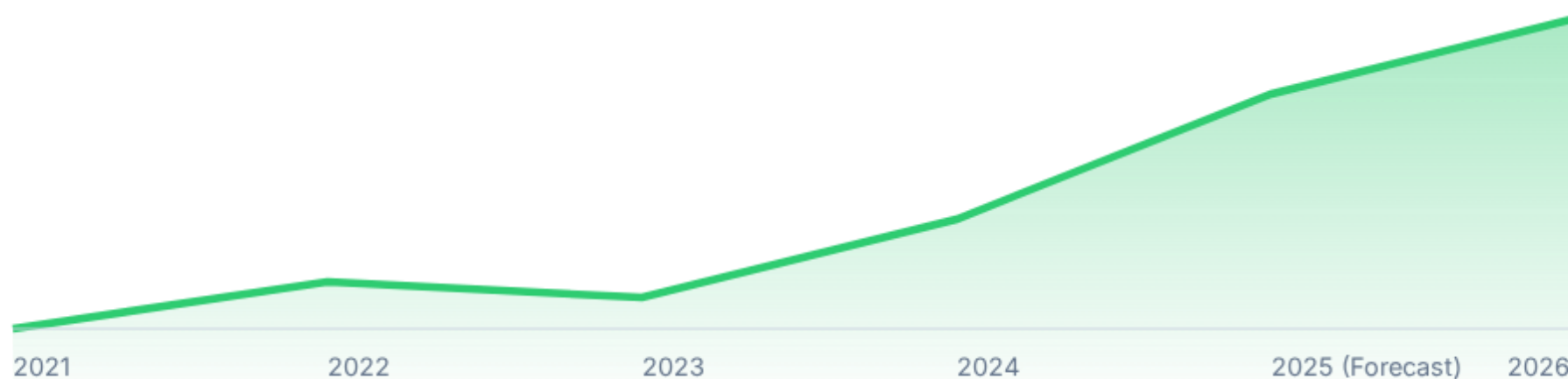
The 1,500 MW Anomaly

Unaccounted-for data dropouts represent severe multi-million dollar revenue leaks and NERC CIP compliance risks.

Project Sentinel was designed specifically to isolate these anomalies from ambient environmental noise and cyclical demand fluctuations.

² **Source:** PJM Interconnection Base Residual Auction Results; PJM Load Analysis Subcommittee Regional Transmission Planning Data Center Impact Reports.

NEW JERSEY PEAK LOAD FORECAST



Load projections indicate a sharp trajectory increase due to building electrification and data center proliferation.

³**Source:** PJM Interconnection Load Forecast Report; Regional Transmission Expansion Plan (RTEP) Growth Projections for the New Jersey Locational Delivery Area.



PROJECT SENTINEL

System Architecture & Forensic Logic

Framework

CORE SYSTEM CAPABILITIES



Anomaly Isolation

Proprietary forensic logic separates authentic grid stress from telemetry "ghost" signals caused by sensor latency.



NERC Compliance

Automated auditing tracks every data dropout, ensuring absolute alignment with national grid security standards.



Real-time Mitigation

Instantaneous detection of 1,500 MW shifts allows for automated load balancing and revenue protection.

PERFORMANCE COMPARISON

Metric	Legacy Monitoring	Project Sentinel	Business Impact
Dropout Detection Speed	15 - 30 Minutes	< 500 Milliseconds	Critical Risk Reduction
False Positive Rate	12.4%	0.8%	Operational Efficiency
Audit Trail Automation	Manual Reporting	Real-time Immutable Logs	100% Compliance
Unhedged Anomaly Limit	Up to 1,500 MW	Isolated at 50 MW	\$Multi-Million Dollar Revenue Assurance

VISUALIZING THE GRID COMMAND

Forensic Visualization

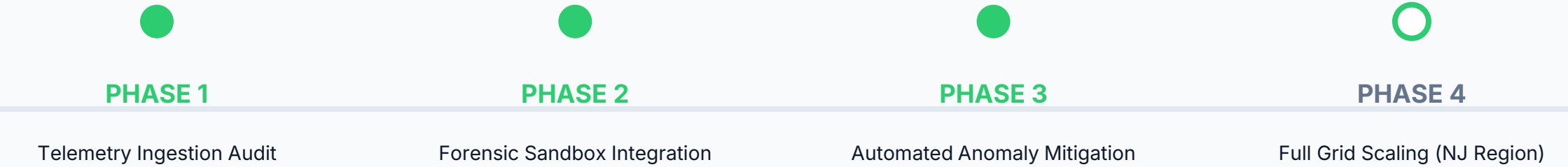
Project Sentinel integrates high-fidelity BigQuery sandboxes to visualize grid stress in real-time. This allows operators to see "through the noise" of raw telemetry data.

By applying Cyber Security by Design, we ensure that the ingestion pipeline is as secure as it is intelligent, meeting the rigorous demands of enterprise-grade infrastructure protection.

"Securing mission-critical systems across enterprise, cloud, and infrastructure."



DEPLOYMENT ROADMAP



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"In a \$10 Billion utility market, a single data dropout is a **multi-million dollar liability**. Project Sentinel transforms this risk into an auditable engine of **revenue assurance**."

— THE EXECUTIVE MANDATE

QUESTIONS?

Secure the Grid. Protect the Revenue.

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IMAGE SOURCES



https://images.stockcake.com/public/b/f/4/bf43e41c-4863-4a36-9755-e4b6bf80d89c_large/futuristic-command-center-stockcake.jpg

Source: stockcake.com